

Leadership in DevOps:-

- ① Building a Devops Culture.
- ② Cross functional Collaboration.
- ③ Stakeholder alignment.
- ④ Strategic decision-making.

① Building a devops Culture

- 1 - Collaboration
- 2 - Communication.
- 3 - Shared Responsibility

→ emphasised s/w dev. & operation team

Key Practices:-

- ① Blameless postmortems to learn from failure.
- ② Continuous learning, and experimentation.
- ③ Implement the automation where ever it possible.
- ④ Align goal across development, QA & operation.

Leadership Role:-

- ① Set the tone by modeling collaborative behaviour.
- ② Invest training & tools that support Devops practices.
- ③ Break down organization silos & encourage shared Accountability.

... collaboration

② Cross-functional Collaboration:-

People across different team (dev, ops, QA, security) work together seamlessly.

Key Practice:-

- ① Create Cross functional team.
- ② Collaboration tool. (e.g. slack, jira, github) to keep everyone Aligned.
- ③ Practice shared planning & retrospective to understand diff. perspective.

Leadership Role:-

- ① Open Communication & mutual respect.
- ② Reduce handoffs by creating integrated workflow.
- ③ Recognize & reward team effort not just individual performer.

③ Stakeholder Alignment

Key Practice:-

- ① Communicate the business value.
- ② Shared goals & metrics.
- ③ Hold regular checkin with stakeholders to track progress & adopt priorities.

Leadership Role:-

- ① Bridge b/w technical & non-technical teams & entire Business Team.

- ① Bridge between -
- ② Devops → Strategic Business Team.
- ③ Manage expectation & Buy in across the org.

④ Strategic Decision Making:-

Key Practices:-

- ① Use Metrics & observability Tool
- ② Buy vs Build
- ③ Evaluate & adopt emerging technology

Leadership Role:-

- ① Business goal & Timeline.
- ② Proactive Risk Management.
- ③ Innovation while maintaining system stability & Customer satisfaction.

Question:

A cloud infrastructure team must provision **identical environments** across **multiple regions** with **consistent configurations**, regardless of current states.

They must also apply **time-sensitive updates** efficiently.

Considering these requirements, which **Infrastructure as Code (IaC)** approach best ensures **consistency** and **automated enforcement** across all target regions?

Options:

- ✓ A. Declarative IaC ensures consistent state enforcement across all regions effortlessly.
- B. Imperative IaC provides greater flexibility for handling specific regional variations initially.
- C. Either approach is equally suitable if the team has sufficient expertise in both methods.
- D. Imperative IaC allows direct state manipulation to reflect current regional differences promptly.

... consistency.

Question:

A sysadmin finds that a production database server's actual setup differs from Terraform's state due to manual changes.

To fix this configuration drift and keep Terraform state accurate, what approach should be used to synchronize the state and infrastructure correctly?

Options:

- A. Manually edit the Terraform state using `terraform state push` to reflect the changes made.
- B. Destroy and recreate the database server using Terraform, which discards existing data and setup.
- C. Import the existing database server into Terraform and generate the configuration from scratch.
- ☒ D. Use `terraform refresh` to update the state, then apply the changes via `terraform apply`.

inconsistent

bypass the validation & corrupt the state

loss data

update the state

overwrite the manual changes

Question:

A development team is launching a new microservice for processing user data. They want to ensure reliability by selecting the most important SLI to monitor. Which SLI best reflects the service's accessibility and uptime?

Options:

- A. Request success rate measures how often requests are handled well.
- ☒ B. Service availability measures the percentage of time the service functions correctly.
- C. Average request latency gauges typical response time for user calls.
- D. Number of processed transactions indicates throughput and workload capacity.

Question:

A system experienced a minor outage after a routine software release on Friday morning. The team decides to start the root cause analysis process to identify the issue and prevent recurrence. What should be the first step in their investigation?

Options:

- A. The team immediately implements the fix suggested by developers.
- B. The team assigns blame to developers for the system error.
- ☒ C. Gather all data and information related to the incident thoroughly.
- D. Skip data gathering and apply previously known fixes immediately.

Question:

During a large-scale DevOps transformation, a critical database update requires immediate deployment. The engineering team validated it, but compliance and security stakeholders have concerns about audit risks. Considering the tight schedule and urgent business needs, what is the best approach for a DevOps Leader to manage this situation effectively?

Options:

- ☒ A. Facilitate a meeting with engineering, security, and compliance to align goals and address concerns.

- B. Deploy immediately and inform compliance and security teams after the update.
- C. Postpone deployment indefinitely until all stakeholder concerns are fully resolved.
- D. Escalate the matter to executive management for a final decision on deployment urgency.

Question:

An application development team builds a cloud-native application. The DevOps team wants to deploy it via Git using the principles of the **GitOps approach**.

As per the GitOps process, a developer pushes the application's code into the Git repository and notices the merge code changes in the **Trunk branch** made by another developer.

Which is the **correct GitOps principle** for deploying the application?

Options:

- A. Defining the whole application in a declarative form using the YAML file. ✗
- ☒ B. Using Git as the multiple source of truth and versioning the canonical desired system/environment state in Git.
- C. Approving changes to the desired state manually. ✗
- D. Employing Infrastructure as service (IaS) and ensuring correctness. ✗

Question:

A development team has implemented CI/CD pipelines with policy-based deployments using a central governance tool. To ensure compliance and security, the governance team requires comprehensive audit trails of all pipeline activities including approvals, policy evaluations, and deployment outcomes.

The team faces a challenge in consolidating audit logs from various pipeline stages and enforcing consistent logging formats across different toolsets.

What action should the team perform to enhance audit logging?

Options:

- A. Collect logs but keep them in the native tool formats.
- B. Disable verbose logging to reduce storage and processing of audit data.
- C. Archive logs without any further analysis or real-time monitoring processes.
- ☒ D. Implement a centralized logging system to standardize collection and consolidate audit logs across pipelines.